

FROM OFFLINE FUSION TO ONLINE DEPLOYMENT: ANOMALY DETECTION ON A UR10e COBOT

Adı SOYADI^{1*}, Adı SOYADI² (Dergi editörlüğü tarafından basım aşamasında yazılacaktır.)

1 Yazar 1 Adresi, ORCID No : <https://orcid.org/>

2 Yazar 2 Adresi, ORCID No : <https://orcid.org/>

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Abstract

Early detection of anomalies in collaborative robots matters for operator safety and production continuity. Previous work introduced a score-level fusion of a physics-informed residual autoencoder and a data-driven raw signal autoencoder for a UR10e cobot, evaluated entirely offline. This paper takes that framework as its starting point, extends it in four directions and delivers it as a detector running on the physical cell. The residual definition is completed with a friction term fitted outside the validated inverse-dynamics solver, removing 87 % and 92 % of the residual spread at the wrist joints. The evaluation is placed on a run-disjoint footing with five training seeds, on which the fusion retains its advantage over both single models (F1 0.791 against 0.627 and 0.614). A physically consistent injection protocol, in which faults perturb only the measured channels and the residual is recomputed, reduces the fusion margin from +0.189 to -0.003 PR-AUC in every seed; the complementarity is traced to one injected amplitude fifty-one times smaller than its physical propagation. The detector was implemented as a 500 Hz ROS 2 node whose feature engine matches the offline pipeline to floating-point rounding. On hardware the offline-derived threshold did not transfer, because both models operate far outside their training distribution and a five-per-cent-weighted model supplied 29 % of the fused score. After recalibration on the cell the detector caught three operator-confirmed collisions with no false alarm, while a low-amplitude contact remained below normal motion.

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ÇEVİRİMDIŞI BİRLEŞİMDEN ÇEVİRİMİÇİ GERÇEKLEMeye UR10e ANOMALİ TESPİTİ

Anahtar Kelimeler

Anomali tespiti
İşbirlikçi robotlar
LSTM özkodlayıcı
Skor birleşimi
Gerçek zamanlı sistem

Öz

İşbirlikçi robotlarda anomalilerin erken tespiti operatör güvenliği ve üretim sürekliliği açısından kritiktir. Önceki çalışmada, bir UR10e kobot için fizik tabanlı kalıntı özkodlayıcısı ile veri güdümlü ham sinyal özkodlayıcısının skor düzeyinde birleşimi sunulmuş ve yalnızca çevrimdışı değerlendirilmişti. Bu makale o çerçeveyi başlangıç noktası olarak dört yönde genişletmekte ve gerçek hücrede çalışan bir dedektör olarak teslim etmektedir. Kalıntı tanımı, doğrulanmış ters dinamik çözücünün içermediği sürtünme terimiyle tamamlanmış, bilek eklemlerinde kalıntı yayılımı %87 ve %92 azalmıştır. Değerlendirme koşu-ayrık bölme ve beş eğitim tohumu üzerine oturtulmuş, bu koşulda birleşim iki tekil modeli de geçmiştir (F1 0,791'e karşılık 0,627 ve 0,614). Arızaların yalnız ölçülen kanallara uygulandığı ve kalıntının yeniden hesaplandığı fiziksel olarak tutarlı bir enjeksiyon protokolü ise birleşim marjını her tohumda +0,189'dan -0,003 PR-AUC'ye indirmektedir; tamamlayıcılığın kaynağı, fiziksel yayılımından elli bir kat küçük seçilmiş tek bir enjeksiyon genliğidir. Dedektör, öznelik motoru çevrimdışı hatla kayan nokta düzeyinde örtüşen 500 Hz'lik bir ROS 2 düğümü olarak gerçekleştirilmiştir. Donanımda çevrimdışı türetilen eşik taşınmamış; sebebi her iki modelin de eğitim dağılımının çok

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dışında çalışması ve %5 ağırlıklı modelin birleşik skorun %29'unu sürüklemesidir. Hücrede yeniden kalibrasyondan sonra dedektör, operatörce doğrulanmış üç çarpışmayı yanlış alarm üretmeden yakalamış, düşük genlikli bir temas ise normal hareketin altında kalmıştır.

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1. Introduction

Collaborative robots share their working volume with human operators, so anomalies such as a collision, a degrading motor or a failing sensor have to be detected before they become incidents. Two families of methods dominate. Physics-based residual methods compare a dynamic model of the robot with the measured forces and treat the difference as a fault indicator; they reduce dimensionality and suppress noise, but they also suppress low-amplitude sensor anomalies. Data-driven methods learn complex patterns directly from high-dimensional raw signals, but they lack the structure a physical model provides. The two families therefore fail on different fault types, and that asymmetry can be exploited by fusing them.

In earlier work we proposed exactly such a fusion for a UR10e cobot (Yılmaz, Kahraman, Yılmaz, Yavuz and Yayan, 2026). A residual Long Short-Term Memory (LSTM) autoencoder operating on twelve intrinsic and extrinsic residual channels derived from a Functional Mock-up Unit (FMU) inverse dynamics model was combined, at score level, with a raw LSTM autoencoder operating on twenty-four direct sensor channels. That study was evaluated entirely offline on a recorded dataset, and its own future-work list named a single most important open item: running the system online and simultaneously with the robot.

This paper closes that item. The framework of that study is taken as the starting point and extended in four directions, and the result is delivered as a detector running on the physical cell. The residual definition is completed with the friction term the validated inverse-dynamics solver does not contain; the evaluation is placed on a run-disjoint footing with five training seeds so that every number carries an uncertainty; the fault injection is made physically consistent, which turns out to change what the fusion is worth; and the normalisation and threshold are given a form that can be estimated without seeing the

future, then re-estimated on the cell when the offline values prove not to transfer. The detector was then commissioned and validated against provoked collisions.

The gap this study fills is therefore not a new architecture. Fusion frameworks for robot anomaly detection are numerous and are almost always evaluated on recorded data with injected faults. What is rarely reported is which parts of such a framework survive contact with hardware: whether the operating threshold transfers, whether the normalisation is even causal, whether the measured fusion advantage depends on how the faults were injected, and what the detector can and cannot see once it is running. This paper reports those measurements for one complete system, and states the resulting working envelope from both sides.

The contributions are:

- a residual definition completed with a Coulomb-plus-viscous friction term fitted outside the validated solver, which removes 87 % and 92 % of the residual spread at the two wrist joints where the inverse dynamics previously contributed almost nothing;
- a run-disjoint re-evaluation with five training seeds, so that the fusion advantage is reported with an uncertainty rather than as a single number;
- a physically consistent fault-injection protocol, and the measurement that the fusion advantage reported under the inherited protocol disappears under it — traced to a single injected amplitude fifty-one times smaller than its physical propagation;
- a causal formulation of the fusion normalisation and an operating threshold for the fused score, neither of which the offline method defines, together with the procedure that re-estimates both on the cell;
- a real-time ROS 2 implementation whose feature engine is numerically identical to



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the offline pipeline, friction term included, with a measured latency budget;

- a commissioning report in which the offline-derived threshold is shown not to transfer, the cause is traced to a normaliser rather than to the models, and the deployed detector is validated against operator-confirmed collisions with the sensitivity floor stated explicitly;
- an audit of the data path carried out before the extension, whose findings are corrected here and whose effect on the previously published figures is documented in Section 4.5.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the platform and the data, the extended pipeline and the online implementation. Section 4 presents the offline and the real-cell findings. Section 5 discusses them and states the working envelope and the limitations, and Section 6 concludes.

2. Literature Review

Physics-based detection for manipulators is a mature field. Haddadin, De Luca and Albu-Schäffer (2017) survey collision detection, isolation and identification and establish the residual observer as the canonical tool. Li, Han and Xiong (2020) place a force/torque sensor at the bedplate and detect collisions from the resulting residual, while Zhang, Chen and Zou (2024) use an external torque observer for the same purpose. Katsampiris-Salgado et al. (2024) address high-payload collaborative assembly, where the model error itself becomes the limiting factor. The common weakness of this family is that whatever the model does not represent — friction, payload, joint elasticity — is indistinguishable from a fault, and whatever the model filters well is also filtered away when it is the fault.

Data-driven detection for multivariate time series is surveyed by Darban, Webb, Pan, Aggarwal and Salehi (2024). Reconstruction-based autoencoders are the dominant design: Malhotra, Vig, Shroff and Agarwal (2015) introduced LSTM networks for time-series anomaly detection, and Malhotra, Ramakrishnan, Anand, Vig, Agarwal and Shroff (2016) extended the idea to a multi-sensor encoder-decoder. Park, Hoshi and Kemp (2018) applied an LSTM-based variational autoencoder to robot-assisted

feeding, one of the few studies to report on a physical robot. These methods need no fault labels, but they inherit whatever bias the training distribution carries.

Hybrid approaches attempt to combine the two. Liu et al. (2025) constrain an LSTM autoencoder with mechanism knowledge for pump operations, but do not study the fusion of two separate models' scores. Huang, Chen, Deng and Huang (2024) apply graph attention and a Transformer to multivariate anomaly detection without any physics-based preprocessing stage. Correia, Goos, Klein, Bäck and Kononova (2024) survey online model-based anomaly detection specifically, and identify threshold selection and non-stationary operating conditions as open research challenges — precisely the two problems that this paper measures on hardware.

The transfer of a model calibrated on simulated or recorded data to a physical robot is studied mostly in the control and reinforcement learning literature, where it is known as the reality gap (Zhao, Queralta and Westerlund, 2020). For anomaly detection the equivalent question — does the decision threshold transfer? — is seldom asked, because most studies never define an operating threshold at all, reporting instead the best achievable F1 over all thresholds. That statistic requires the labels it is supposed to predict and therefore does not exist online. The present study makes this gap explicit and closes it with a causal threshold rule, then measures what happens when the rule is applied to real hardware.

3. Method

3.1. Platform And Data Collection

Experiments were carried out on a robotic quality inspection platform consisting of a Universal Robots UR10e collaborative robot (six joints, 12.5 kg payload) mounted on a Festo linear axis, together with an inspection chassis frame, shown in Figure 1. The target application is automated visual quality inspection of automotive body panels. A force/torque sensor (FTS) is mounted at the end effector. The cell runs ROS 2 Humble with MoveIt 2, and the recorded motions were produced by eleven different motion planning algorithms so that the normal operating distribution is not dominated by a single planner.



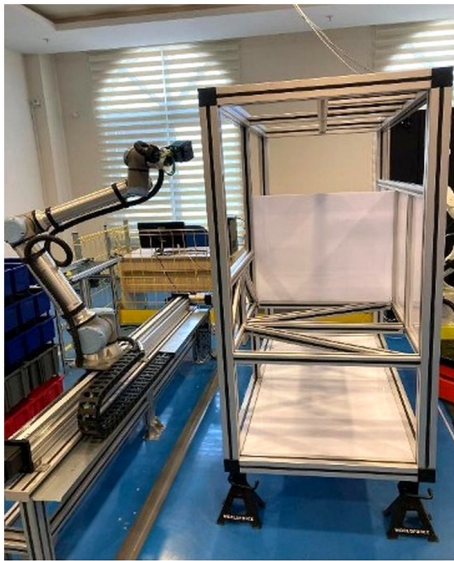


Figure 1. Robotic Quality Inspection Platform
 UR10e robot on a Festo linear axis, with the
 inspection chassis frame on the right.

The source dataset is a 79-day export of joint states and wrench measurements, 1,124,432 samples nominally at 500 Hz. Model training was performed on a laptop with an Intel Core i7-13650HX processor and an NVIDIA RTX 4060 GPU; the deployment workstation is an Intel Core i9-14900 with 64 GB of memory and an NVIDIA RTX 4000 Ada GPU.

The study uses no human subjects and required no ethics committee approval. All measurements were collected on laboratory equipment of the ESOGÜ Intelligent Systems Application and Research Centre, and the principles of research and publication ethics were observed throughout. The anomalies of the commissioning campaign were provoked deliberately by trained operators at reduced speed, with the robot's own protective-stop function active at all times; one of them, reported in Section 4.7, involved an operator's hand being caught between two links, and the protective stop halted the robot without injury.

3.2. Data Investigation And Preparation

The first stage of the audit was the data itself, and it produced the largest single correction. The export is not globally ordered by timestamp. Measured over the 1,124,432 samples, 321,841 consecutive pairs (28.6 %) are discontinuous, 96,107 of them with a negative or zero time step; the median step is the expected 2.00 ms but the range spans from $-5.5 \cdot 10^9$ ms to $+3.6 \cdot 10^9$ ms. Three quarters of the sliding windows (33,700 of 44,974, 74.9 %) therefore splice together two

recording sessions that may be hours or days apart.

This matters twice. It puts an arbitrary jump in the middle of the dynamics an LSTM autoencoder is asked to learn, and it corrupts the joint acceleration. Acceleration is obtained from the velocity channel by a Savitzky-Golay derivative (window 51, order 3) (Savitzky and Golay, 1964), which is a convolution: across a splice it produces a non-physical spike. Since acceleration enters the inverse dynamics directly through the $M(q)\ddot{q}$ term, the error propagates into the residual. Measured on the same 200,000 samples, the standard deviation of $\ddot{q}$ derived across splices is two to six times larger than when the derivative is taken inside continuous runs.

The correction is to segment the export into physically continuous runs. Two consecutive samples belong to the same run only if the time step lies in $(0, 8 \text{ ms}]$ and the joint displacement stays below what the joint velocity limit allows over that step; the second condition catches gaps that a plausible-looking timestamp hides. Runs shorter than 150 samples are discarded, and each run is resampled onto a uniform 2 ms grid. Table 1 gives the outcome. The reduction in sample count is not a loss: the discarded samples are exactly those between which no physical continuity exists.

Table 1. Outcome Of Data Preparation.

Quantity	Value
Raw exported samples	1,124,432
Physically continuous runs	600
Prepared samples	342,480 (685 s of robot time)
Interpolated samples	5.02 %
Run length (median / min / max)	227 / 155 / 35,860

3.3. Current-To-Torque Calibration

The UR ROS 2 driver writes the motor current, in amperes, into the effort field of the joint state message; the field is filled directly from the actual current reported by the controller. The inverse dynamics model, in contrast, produces newton metres. A conversion coefficient is therefore required before the residual of Equation (2) can be formed at all.

In a first rebuild this coefficient was obtained by regressing measured current on model torque, and verified by repeating the same regression. That verification is circular: the



coefficient is chosen so as to satisfy the equality that is then tested. The coefficient was therefore re-derived from a source independent of the residual under test. When the robot is nearly at rest ($|\dot{q}| < 0.005$ rad/s) the joint torque is dominated by gravity, so

$g(q) \approx k \cdot (i - i_0)$ with $g(q)$ taken from the closed-form gravity vector rather than from the model being audited. Of 197,090 quasi-static samples, 65,697 entered the regression. Table 2 lists the result.

Table 2. Measured Current-To-Torque Coefficients And Their Sensitivity.

Joint	Nm/A	R ²	Meas.	$\rho -50\%$	$\rho +50\%$
shoulder_pan	10.522	—	no	0.968	0.995
shoulder_lift	10.522	0.984	yes	0.162	0.637
elbow	9.130	0.993	yes	0.023	0.758
wrist_1	3.409	0.000	no	0.890	0.968
wrist_2	3.409	0.012	no	0.957	0.993
wrist_3	3.409	0.010	no	0.994	0.999

R² is that of the quasi-static gravity regression and “Meas.” marks the joints on which the coefficient could be measured directly. The last two columns give the correlation between the intrinsic residual recomputed with the coefficient perturbed by $\pm 50\%$ and the nominal one, over all 342,480 prepared samples. Coefficients that could not be measured are copied within the same module family or scaled by the torque-limit ratio, and the node warns about them at start-up.

The consequence of the four assumed coefficients was measured rather than argued. Each coefficient was perturbed by $\pm 50\%$ and the intrinsic residual recomputed, which requires no further evaluation of the inverse dynamics because the model torque is stored separately from the measurement. On the four joints whose coefficient could not be measured, the perturbation leaves the residual almost unchanged in shape: the correlation with the nominal residual is between 0.890 and 0.999 and the effect is close to a pure rescaling, with the standard deviation changing by factors of 0.55 to 1.45. Since the autoencoders standardise their inputs, a transformation of that form cannot alter the learned dynamics. On the two joints whose coefficient was measured, the same perturbation is destructive: the correlation falls to 0.162 for shoulder_lift and to 0.023 for the elbow at -50% . The asymmetry is itself diagnostic. Where the current term carries the gravity torque that the model is meant to cancel, mis-scaling it destroys the cancellation; where the model contributes little, as at the wrists, the coefficient acts as a scale factor only. The assumed coefficients therefore affect the physical interpretation of a fault amplitude in newton metres rather than the detection performance.

3.4. Hybrid Residual Decomposition

The inverse dynamics model packages the Newton-Euler formulation of the UR10e in the Functional Mock-up Interface format (Blochwitz et al., 2011):

$$\hat{\tau}_{model} = M(q) \cdot \ddot{q} + C(q, \dot{q}) \cdot \dot{q} + g(q) \quad (1)$$

The total residual is the difference between measurement and model,

$$r_{total} = \tau_{meas} - \hat{\tau}_{model} \quad (2)$$

the extrinsic residual is the projection of the end-effector wrench into joint space,

$$r_{ext} = J(q)^T \cdot F_{FTS} \quad (3)$$

and the intrinsic residual is the difference of the two,

$$r_{int} = r_{total} - r_{ext} \quad (4)$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})$ the Coriolis and centrifugal matrix, $g(q)$ the gravity vector, $J(q)$ the geometric Jacobian, τ_{meas} the measured joint torque and F_{FTS} the measured wrench. This decomposition makes r_{int} sensitive to internal faults such as motor degradation or an encoder glitch, and r_{ext} sensitive to external interaction such as a collision or a payload change. Because Equation (4) is a definition, the numerical consistency of the decomposition can be tested directly: over 200,000 samples the identity holds to a residue of $7.1 \cdot 10^{-15}$, and the analytical Jacobian agrees with a numerical one to machine precision.

Equation (3) requires an agreement of reference frames whose violation produces no diagnostic. The UR e-Series driver rotates the wrench into the tool frame and publishes it with the tool frame identifier, whereas $J(q)$ is defined in the base frame. Multiplying the two directly produces a wrong residual with no



error message. Measured over 40,000 samples, the correlation between the correct and the unrotated extrinsic residual is -0.995 on the first joint — the sign inverts — and -0.143 across all channels; on the second joint the amplitude triples. Both force and moment halves are therefore rotated into the base frame before the Jacobian transfer, in the offline pipeline and in the online engine through the same code path.

Two properties of the solver were measured and are not corrected inside it. Over 500 random joint poses the inertia matrix is asymmetric in every pose and its symmetric part is not positive definite, the friction vector is identically zero and no payload model is present. The solver had been validated against the physical robot and is treated as fixed. Its cost is therefore measured rather than hidden: on wrist_2 and wrist_3 the standard deviation of the model torque is only 10 % and 2 % of the measured torque (0.095 and 0.030 Nm against 0.959 and 1.246 Nm), so in those two channels Equation (2) reduces to the measurement itself.

The absent friction term is added outside the solver rather than inside it, which leaves the

validated model untouched while restoring a component that is large in a geared manipulator. Equation (2) becomes

$$r_{total} = \tau_{meas} - \hat{\tau}_{model} - \hat{\tau}_f(\dot{q}) \quad (2a)$$

with a two-parameter Coulomb-plus-viscous term per joint,

$$\hat{\tau}_f = F_c \cdot \tanh(\dot{q} / \varepsilon) + F_v \cdot \dot{q} \quad (2b)$$

where $\tanh$ replaces the sign function because $\text{sign}(\dot{q})$ is discontinuous at zero crossings and this robot spends most of its time at low speed; a sharp sign injects a step into the residual at every reversal, which is precisely the shape a detector is meant to flag. The smoothing width is $\varepsilon = 0.02$ rad/s. Coefficients are fitted by robust least squares (three iterations of 3σ clipping) on moving samples only and, decisively for the evaluation, exclusively on the training runs of the split defined in Section 3.7. Fitting them on all data would make the correction itself a leakage channel. Table 3 lists the result.

Table 3. Friction Coefficients, Fitted On Training Runs Only.

Joint	F_c [Nm]	F_v [Nm·s/rad]	R ²	$\sigma(r)$ before	$\sigma(r)$ after	Δ
shoulder_pan	7.65	35.76	0.764	13.20	6.09	−54 %
shoulder_lift	8.78	26.19	0.799	11.91	4.98	−58 %
elbow	5.53	19.97	0.896	7.66	2.46	−68 %
wrist_1	0.95	1.29	0.379	1.41	0.75	−47 %
wrist_2	1.51	1.58	0.947	1.53	0.20	−87 %
wrist_3	1.38	2.05	0.974	1.90	0.15	−92 %

$\sigma(r)$ is the standard deviation of the total residual on moving samples ($|\dot{q}| > 0.02$ rad/s) before and after the term is subtracted. The regression absorbs any velocity-correlated model error, not friction alone; these are not claimed to be the true tribological coefficients. The effect is largest exactly where the inverse dynamics contributes least: on wrist_2 and wrist_3 the residual loses 87 % and 92 % of its spread, so the admission above — that those channels carry no model information — no longer holds once the term is included. Whether that improves detection is a separate question, answered in Section 4.4.

The online feature engine applies the identical expression from the same coefficient file, and the equivalence test of Section 3.9 is run with the term active.

3.5. Dual LSTM Autoencoders

Both models share a symmetric encoder-decoder architecture that compresses a sliding window of $T = 100$ samples (0.2 s) into a latent vector. A two-layer LSTM encoder reduces the input sequence to a fixed-size latent vector; the decoder repeats it over T steps and reconstructs it with symmetric LSTM layers. Dropout of 15 % is applied between layers. Both models are trained on

normal operating data only, with a mean squared error loss, and the anomaly score of a window is its reconstruction error. Each model's own threshold is the 97th percentile of the reconstruction errors on the validation set.

Windowing uses a stride of 25 samples, that is 75 % overlap. Two rules are enforced jointly: no window may span two runs, and windows containing samples invalidated by the derivative edge margin are dropped. Training and validation windows come from disjoint sets of physical runs (Section 3.7), so no sample is shared between them. Each model is trained five times with different seeds;



every offline figure in this paper is reported as a mean over those five runs with its standard deviation, because a single training run cannot separate a difference between methods from the noise of initialisation. Table 4 reports the architectures and the training outcome.

Table 4. Model Architectures And Training Outcome (five seeds, mean $\pm$ s.d.).

	Residual AE	Raw AE
Input channels	12	24
Hidden / latent	128 / 32	256 / 64
Parameters	478,892	1,907,032
Train / val. windows	6,756 / 2,162	6,756 / 2,162
Best epoch	86 $\pm$ 41	83 $\pm$ 42
Best validation loss	0.092 $\pm$ 0.016	0.036 $\pm$ 0.005
Threshold θ (P97)	0.887 $\pm$ 0.412	0.412 $\pm$ 0.038

Optimiser Adam ($\text{lr} = 10^{-3}$, $\beta = 0.9/0.999$), batch size 256, gradient clipping at 1.0, ReduceLROnPlateau (factor 0.5, patience 8), early stopping with patience 25 over at most 300 epochs; each model trains in under a minute on the RTX 4060 laptop. Both validation losses are far below those of the reference study (0.181 and 0.320), and the reason is the composition of the validation set rather than a better model: the sixteen validation runs were chosen to lie inside the training joint range (Section 3.7), so validation resembles training more closely than an index cut does. The friction term does not explain it — the raw model never sees the residual channels and its loss is 0.036 with and without the term, while the residual model's loss is 0.092 with it against 0.062 without. A validation set that resembles training gives a narrow distribution, and θ is a percentile of that distribution; Section 4.7 measures what this costs on hardware. Across seeds the residual threshold varies by 46 % of its mean, the raw one by 9 %. Values here are five-seed means; Section 4.7 quotes the deployed seed, whose thresholds are 1.3723 and 0.3934.

3.6. Score-Level Fusion And Causal Normalisation

The normalised scores of the two models are combined by a weighted average,

$$S_{\text{fused}} = w_{\text{res}} \cdot z_{\text{res}} + w_{\text{raw}} \cdot z_{\text{raw}} \quad (5)$$

with $w_{\text{res}} + w_{\text{raw}} = 1$. The weight is not tuned here: $w_{\text{res}} = 0.95$ is adopted a priori from the reference study and no evaluation set is consulted to choose it. This matters because the reference study selected that value as the argmax of Best F1 over its own test set; fixing it removes the selection entirely rather than moving it to another set. The sweep over $[0, 1]$ in steps of 0.05 is still run, but Section 4.3 reports it as a sensitivity analysis, not as a search. The reference study normalises each score by a min-max transform whose bounds come from the fault-

injected test set. That is not usable online for two reasons: the bounds contain future maxima, and those maxima are never seen in normal operation. Measured on that study's own test set, where the maxima are 4341 and 7752, the entire normal operating range compresses to the order of 10^{-4} , the fused threshold becomes stricter than either single threshold, and motor drift and encoder glitches are never detected at all.

Both normalisations are affine, so the online engine implements a single code path,

$$z = (S - lo) / span \quad (6)$$

and the bounds are taken from clean validation windows rather than from the test set. This keeps the published formula and reduces the deviation from the reference method to a change of bounding set only. The fused operating threshold extends the study's own rule to the fused score: the 97th percentile on clean validation windows. Offline this gives the expected 3 % false-alarm rate. Section 4.7 shows that neither the bounds nor the threshold survive the move to hardware, and Section 3.11 gives the procedure that replaces them there.

3.7. Data Split, Fault Injection And Evaluation Protocol

The 600 physical runs are partitioned once, and every consumer reads the same partition: the friction fit of Section 3.4, the autoencoder training of Section 3.5, and the evaluation below. The partition is by run, not by row index. The reference study cut the recording at a fixed fraction and then built its test set from every window of the recording; because training used only the first part, four fifths of its evaluation windows were windows the models had already reconstructed. Measured on that arrangement, the reconstruction error of clean windows inside the training range was 0.007 against 0.486 outside it, a factor of seventy for both models alike.

The run lengths make a naive split unusable: 521 of the 600 runs are shorter than one second, while the 32 longest carry 54 % of the samples and the single longest carries 10.5 %. Runs are therefore allocated in two passes. Runs long enough to host a fault are distributed by count so that each split receives its own independent fault events; the remaining short runs then fill the sample quota. Table 5 gives the outcome. Joint-range



coverage was checked afterwards: validation

and test lie inside the training range on all six joints.

Table 5. Run-Disjoint Split.

Split	Runs	Samples	Share	Windows	Fault hosts
Train	489	235,657	68.8 %	6,756	74
Validation	16	56,244	16.4 %	2,162	16
Test	95	50,579	14.8 %	1,504	16

Verified after allocation: zero samples are shared between any two splits once window overlap is accounted for, and no window reaches into a run belonging to another split. The clean-window score ratio between test and training falls from the seventy of the reference arrangement to 0.48.

Ground-truth labels do not exist, so four synthetic fault scenarios are injected, listed in Table 6. Joint 3 (elbow) and joint 5 (wrist 2) are chosen because they carry the largest gravity load and the widest motion range. Faults are injected into every fault-hosting run of the target split separately, so each scenario contributes 16 independent events to the test set. Under the reference protocol the mask was a single contiguous block over the concatenated recording, which gave one event per scenario and, worse, tied each scenario to a fixed position in the recording. A window is labelled anomalous when at least 10 % of its samples overlap the fault mask.

Table 6. Synthetic Fault Scenarios.

Scenario	Affected channel	Amplitude	Extent
Motor drift	Joint 3 torque, linear ramp	15 Nm	30–38 %
Collision	All FTS channels, Gaussian pulse	30 N	centre 65 %
Encoder glitch	Joint 5 position, step	1.5 rad	from 92 %
Sensor noise	All FTS channels, Gaussian noise	3.5 N	15–23 %

Extent is expressed as a fraction of the run the fault is injected into. The amplitudes above are those of the reference study and apply to the measured channels.

Two injection protocols are evaluated. The inherited protocol perturbs each representation space separately with its own amplitude, so the residual model receives a hand-chosen perturbation rather than one derived from the measurement. The physical protocol perturbs only the measured channels and recomputes the residual through the pipeline itself, so the disturbance in the residual is whatever the inverse dynamics and the Jacobian transfer produce. The calibration offset is frozen from the clean pipeline when the residual is recomputed; refitting it on injected data shifts the residual of untouched runs as well. Section 4.4 reports both protocols, because they do not agree.

3.8. Verifying Where The Earlier Figures Came From

Before extending a result it is worth establishing what the earlier numbers measured. The measurement-level checks of Sections 3.2 to 3.4 point at a consistent arrangement, and that arrangement is testable. The hypothesis is that the previously published figures come from a pipeline with mixed units: τ_{meas} in amperes, $\hat{\tau}_{\text{model}}$ in newton metres, r_{ext} in newton metres and computed with an unrotated wrench, the three subtracted directly, and fault amplitudes specified in newton metres added as bare numbers to ampere channels. A further finding compounds it: the residual generator zeroes the model torque on joints whose fit quality falls below a threshold, and under mixed units that guard fires on all six joints, silently reducing the residual to the measurement itself without any message.

The hypothesis makes a falsifiable prediction: rebuilt in that configuration, the signature findings of the reference study should return — specifically the blindness of the residual model to sensor noise and the roughly one-quarter share of anomalies detected by the residual model alone. The reproduction is scripted and was rerun for this paper; its result is reported in Section 4.5.

3.9. Online Implementation

The online system differs structurally from the offline pipeline in three ways, and its design is entirely the management of those three differences. Causality: nothing computed over a whole dataset may be used live. Time budget: at 500 Hz there are 2 ms per sample and, with a stride of 25, 50 ms per decision. Continuity: the offline pipeline may segment the data into runs, whereas the live system notices an interruption only after it has happened and must reset its state. Figure 2 shows the resulting data path.



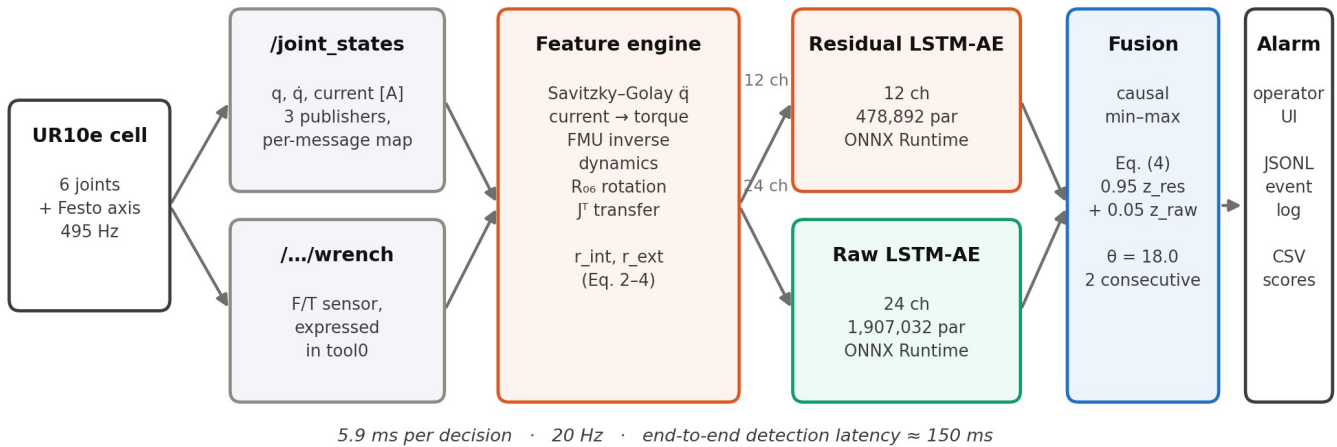


Figure 2. Data Path Of The Online Detector every stage between the robot and the alarm runs once per decision, twenty times a second.

The feature engine is the sample-by-sample equivalent of the offline computation. It keeps a 51-sample ring buffer and, for every new sample, emits the features of the sample at the centre of the buffer: the Savitzky-Golay derivative applied as an inner product with precomputed coefficients, the current-to-torque conversion, the inverse dynamics evaluation, the base-frame rotation and the Jacobian transfer. The centred derivative costs a structural delay of 25 samples (50 ms), which is the price of a noise-free acceleration and is identical to the offline definition.

Equivalence was verified rather than assumed. The prepared dataset was pushed through the online engine sample by sample and compared with the offline feature file: over 8 runs and 61,524 samples the largest absolute differences are $1.4 \cdot 10^{-14}$ for torque, exactly zero for model torque, $8.0 \cdot 10^{-15}$ for the total residual, $2.7 \cdot 10^{-15}$ for the extrinsic and $8.9 \cdot 10^{-15}$ for the intrinsic residual. These are floating-point rounding levels, so the deployed node sees the features the models were trained on, not merely similar ones.

Two optional rules were added on top of the reference method. An adaptive rule fires when the fused score exceeds median + $k \cdot 1.4826 \cdot \text{MAD}$ of the last 600 decisions, where $1.4826 \cdot \text{MAD}$ is a robust estimate of the standard deviation (Leys, Ley, Klein, Bernard and Licata, 2013); it exists because an absolute threshold cannot see a fault that raises the score a hundredfold inside a run whose baseline sits far below that threshold. The scale is floored so that a nearly constant baseline does not make the rule hypersensitive, the baseline is frozen while an

alarm lasts, and the freeze expires after 3 s so that a long alarm is treated as a regime change rather than as an attack. The second rule requires two consecutive decisions (100 ms) before an alarm is raised. Both can be switched off with a single parameter, which restores the reference behaviour exactly.

The detector is packaged as a ROS 2 node (Macenski, Foote, Gerkey, Lalancette and Woodall, 2022) that keeps the whole computation in a ROS-independent core; the node is a thin wrapper around it, so replay tests exercise the class the node actually runs. The models are exported to ONNX and executed with ONNX Runtime, with two threads per session so that a greedy inference cannot starve the 500 Hz callback loop. Every decision is written to a comma-separated log and every alarm to a line-buffered JSON-lines event log; these two files are the raw material from which the real-cell thresholds of Section 4.7 were re-measured.

Deployment surfaced three connection-layer measurements, all of them silent. The wrench topic advertised in the controller configuration is not the one the running broadcaster uses, and subscribing to the wrong name yields no wrench and therefore no score at all. The joint state topic carries three different name sets from three publishers, of which the UR data arrives in a seven-element set in scrambled order at about 112 non-UR messages per second; the joint mapping must therefore be resolved per message and cached per name set, since resolving it once and applying it blindly reads five of six channels from the wrong joints. Finally the measured sample rate is 495 Hz



rather than 500 Hz, which the fixed-step derivative tolerates at the one per cent level.

3.10. Operator Interface

The detector was integrated into the laboratory's existing web dashboard as a third tab. A collector subscribes to the node's decision, alarm and score topics, keeps a 60-second ring buffer and pushes it to the browser at 5 Hz. Each collector runs its own single-threaded executor: in a first version both collectors used the shared global executor, two threads then raced on the same wait set, the node disappeared from the graph, the subscriptions received nothing and no error was printed.

Three interface decisions follow directly from measurements rather than from taste. Alarms are latched on the rising edge and held for five seconds, because the clearest verified real event of the campaign lasted 0.25 s — five decisions — and level sampling at 5 Hz misses it entirely; in a controlled test with synthetic 0.25 s alarms, edge capture saw four events out of four and level sampling only two. The vertical axis is linear and framed on the live threshold rather than on a fixed range, because the score scale changes when the detector is recalibrated and a fixed frame leaves the whole trace flat against one edge; a logarithmic axis was rejected because it visually flattens the region above the threshold, which is the only region that matters. In the event table the peak score is the primary column and the entry score is secondary, because an entry value read without its peak led to a misjudged event during commissioning — event 3 of Table 14 enters at 13.75 and peaks at 25.04. file that the detector never touches, and they accumulate the labelled set on which Section 4.7 rests. Both the axis and the displayed per-model thresholds are read from the live decision stream rather than compiled in, so a recalibration cannot leave the display describing a configuration that is no longer running.

3.11. Threshold Calibration On The Cell

The offline procedure of Section 3.6 fixes (lo, span) and θ from clean validation windows. Section 4.7 shows that these do not transfer: the models are far outside their training distribution during ordinary operation of the real cell, so a threshold placed at the 97th percentile of an offline distribution sits below the cell's normal range. The deployed system

therefore re-measures the same three quantities on the cell, using the same formula and changing only the set they are estimated from — clean decisions logged on the robot instead of clean validation windows.

Two details are forced by measurement. First, the scale is the 97th percentile of each model's own clean scores, not the min-max range of the reference formula: on the cell the min-max range is set by the payload plateau of Section 4.7, which inflates the residual span to 30.8 and makes an event of ordinary magnitude invisible. Second, the calibration run must cover every task the cell performs. A calibration taken on the inspection cycle alone left the pick-and-place task outside its range and the detector alarmed on normal transport. The deployed configuration is calibrated on 15,627 clean decisions spanning both tasks.

4. Findings

4.1. Offline Performance Of The Extended Pipeline

The extended pipeline is evaluated on 6,016 test windows of which 553 (9.2 %) are anomalous, drawn from 95 runs that contributed nothing to training, to the friction fit, or to the choice of any threshold. Every figure below is a mean over five training seeds with its standard deviation. Table 7 gives the result and Figure 3 shows the corresponding curves.

Table 7. Overall Performance On The Run-Disjoint Test Set (five seeds, mean $\pm$ s.d.).

Model	AUC	PR-AUC	Best F1
Residual LSTM	0.878 $\pm$ 0.005	0.653 $\pm$ 0.009	0.628 $\pm$ 0.007
Raw LSTM AE	0.870 $\pm$ 0.008	0.668 $\pm$ 0.009	0.697 $\pm$ 0.001
Fusion (0.95/0.05)	0.969 $\pm$ 0.004	0.857 $\pm$ 0.009	0.821 $\pm$ 0.006
Fusion MAX	0.966 $\pm$ 0.004	0.853 $\pm$ 0.010	0.784 $\pm$ 0.012
Residual norm threshold	0.758 $\pm$ 0.000	0.485 $\pm$ 0.000	0.495 $\pm$ 0.000
Isolation Forest	0.753 $\pm$ 0.008	0.418 $\pm$ 0.008	0.415 $\pm$ 0.008
One-Class SVM	0.844 $\pm$ 0.001	0.618 $\pm$ 0.001	0.676 $\pm$ 0.000

Figures from the earlier study are not tabulated beside these, because that evaluation used a different and much larger window set under a protocol whose evaluation windows overlapped its training windows; Section 4.5 covers the comparison in one place. That study also lists an OR row with values identical to its MAX row. OR is a decision-level combination: it produces a binary outcome, not a ranking, so AUC, PR-AUC and Best F1 are undefined for it and the row in fact reports the MAX



score. Its operating-point behaviour appears in Section 4.2 instead. Best F1 is the maximum over all thresholds and therefore requires the labels it is meant to predict; it

is listed only for comparability, which is why Section 3.6 defines a causal operating threshold. Baselines are fitted on the clean training runs.

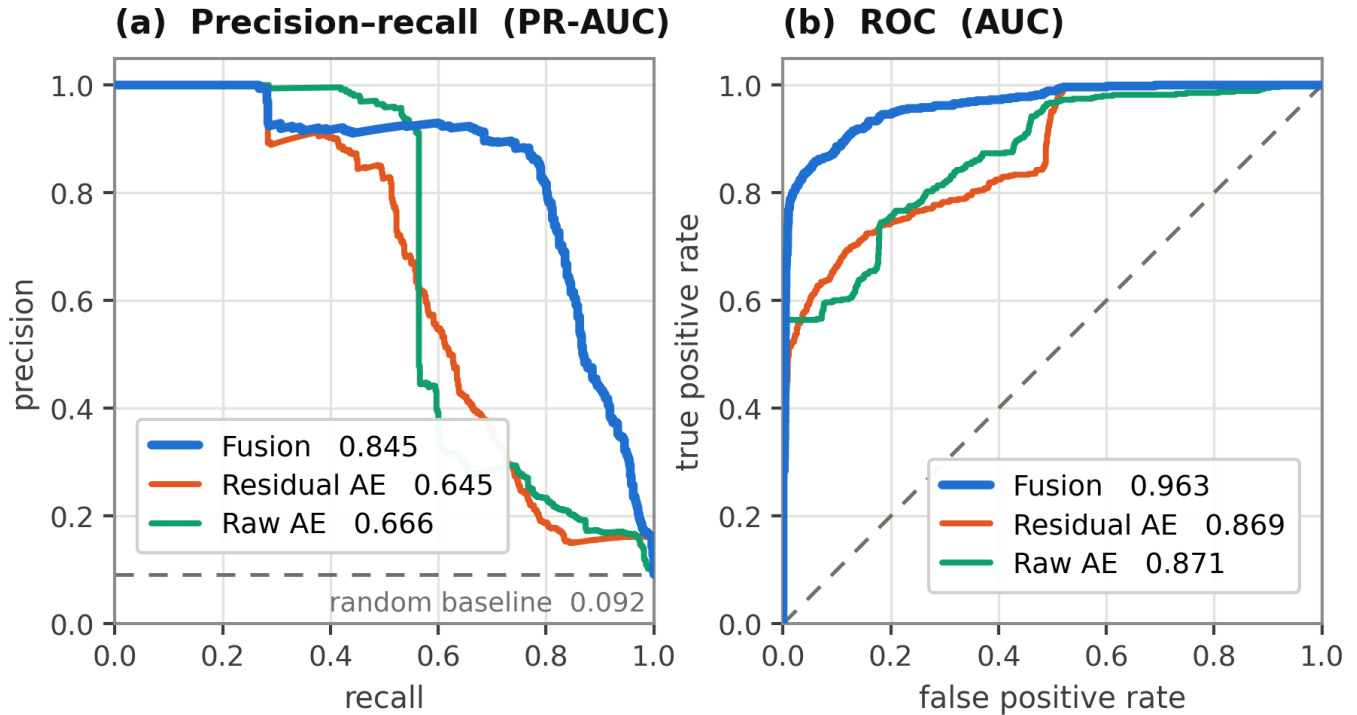


Figure 3. Precision-Recall And ROC Curves On The Run-Disjoint Test Set 6,016 test windows, 9.2 % anomalous, seed 1 of 5. Fusion dominates both single models on both curves; the margin is larger on the precision-recall panel because ROC is optimistic under class imbalance.

At the causal operating point the fused detector reaches precision 0.764 ± 0.086 , recall 0.831 ± 0.055 and F1 0.791 ± 0.023 , against F1 0.627 ± 0.006 for the raw model alone and 0.614 ± 0.016 for the residual model alone. The gain over the better single model is $\Delta F1 = +0.164$. The clean-window score ratio between test and training runs is 0.48 for both models, against the factor of seventy that the reference split produced; the optimism that the earlier arrangement introduced is gone.

The precision spread of ± 0.086 is not noise in the models but in the threshold. Its value across the five seeds varies by 46 % of its mean, because it is the 97th percentile of a

validation distribution that the validation distribution narrowed by the split design of Section 3.7. The ranking metrics, which do not depend on a threshold, vary by less than 0.01.

4.2. Fault-Type Asymmetry And Complementarity

Table 8 breaks the result down by fault type, and reports AUC and Best F1 separately because they are not the same quantity and the distinction changes the reading: on motor drift the residual model has AUC 0.912 but Best F1 0.669, and quoting the second as if it were the first understates the ranking quality by a wide margin.

Table 8. Detection By Fault Type Under Both Injection Protocols (AUC, five seeds).

Scenario	Inherited: residual	Inherited: raw	Physical: residual	Physical: raw
Motor drift	0.912 ± 0.013	0.652 ± 0.014	0.846 ± 0.021	0.652 ± 0.014
Collision	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Encoder glitch	1.000 ± 0.000	0.832 ± 0.041	0.783 ± 0.052	0.832 ± 0.041
Sensor noise	0.682 ± 0.006	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

The raw columns are identical between the two protocols because the raw model reads the measured channels, which both protocols perturb the same way. Only the residual columns move, and they move a great deal — the interpretation is in Section 4.4.



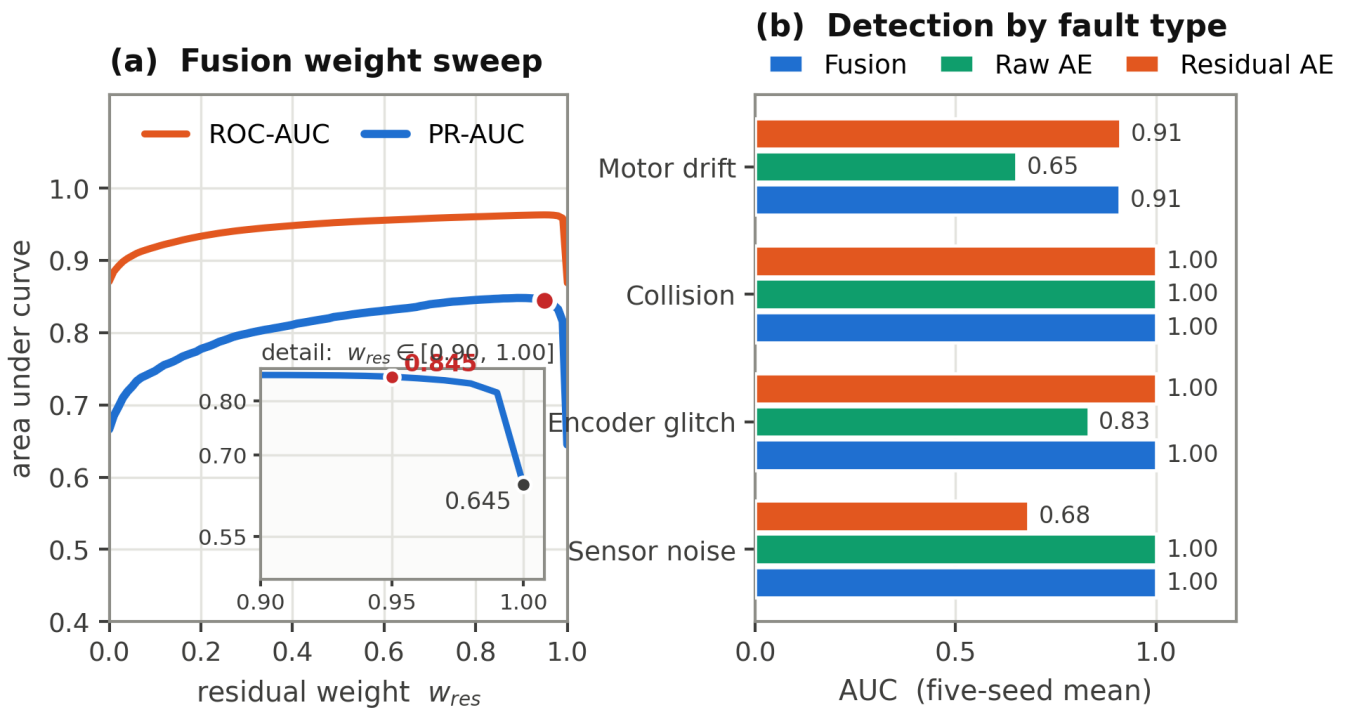


Figure 4. Value Of The Fusion (a) ranking quality against the residual weight, which is fixed a priori at 0.95 rather than selected. (b) detection by fault type under the inherited injection protocol.

Under the inherited protocol the asymmetry that motivates the fusion is present and clean. On the encoder glitch the residual autoencoder is perfect and the raw model reaches 0.832; on sensor noise the roles invert, the residual model falling to 0.682 while the raw model is perfect. At window level, with each model using its own 97th-percentile threshold, 27.6 % of anomalous windows are detected by the residual model only and 28.0 % by the raw model only, against 24.1 % and 25.9 % in the reference study; 15.9 % are caught by neither. The recall of the OR combination is 0.841. Table 9 gives the decision-level ablation.

Table 9. Decision-Level Ablation At The Operating Point (five seeds, mean $\pm$ s.d.).

Configuration	Precision	Recall	F1
Raw only ($w_{res} = 0.00$)	0.704 $\pm$ 0.015	0.565 $\pm$ 0.002	0.627 $\pm$ 0.006
Residual only ($w_{res} = 1.00$)	0.697 $\pm$ 0.111	0.562 $\pm$ 0.058	0.614 $\pm$ 0.016
OR (decision level)	0.701 $\pm$ 0.047	0.841 $\pm$ 0.058	0.762 $\pm$ 0.013
Fusion ($w_{res} = 0.95$)	0.764 $\pm$ 0.086	0.831 $\pm$ 0.055	0.791 $\pm$ 0.023

Inherited injection protocol. The gain over the better single model is $\Delta F1 = +0.164$; the reference study reports +0.161 at a different absolute level and under a protocol with overlapping windows.

4.3. Sensitivity To The Fusion Weight

The weight is fixed a priori at $w_{res} = 0.95$ (Section 3.6), so the sweep below measures sensitivity rather than selecting anything. Figure 4(a) shows a broad plateau over roughly $[0.25, 0.95]$, reproducing the stable region reported by the reference study, with a sharp fall as the raw model is eliminated entirely. Under the inherited protocol the fused PR-AUC at $w_{res} = 0.95$ is 0.857 ± 0.009 against 0.653 ± 0.009 for the pure residual detector, so a five per cent weight on the raw model is associated with $+0.204$ PR-AUC. The margin over the better single model, $+0.189 \pm 0.009$, is positive in every one of the five seeds.

Two independent checks confirm that both models really run. Structurally, every rule of the reference method maps onto the deployed decision path. Operationally, the scoring method of each ONNX session was wrapped and counted over a replay of 6 runs and 5 scenarios: 13,160 decisions produced 13,160 residual-model calls and 13,160 raw-model calls, so neither model is cached or skipped.

4.4. What The Fusion Margin Depends On

The margin reported above is a property of the inherited injection protocol, not of the two representation spaces. The protocol perturbs the residual channels directly with an



amplitude chosen by hand, and one of those choices carries the entire complementarity result. For sensor noise the reference study adds 3.5 N to the wrench and, separately, 0.08 Nm to the extrinsic residual, describing the second as deliberately small so that the scenario 'affects the raw signal only'. Physically, a 3.5 N wrench disturbance propagates into the residual through $r_{\text{ext}} = J(q)^T F$; measured on this robot it produces a standard deviation of 4.1 Nm in the residual channels, fifty-one times the amplitude that was injected by hand. The inverse dynamics does not filter the wrench, it transforms it.

The comparison is not one-sided. For the collision scenario the hand-chosen residual amplitude of 40 Nm agrees with the physical propagation, whose peak is 80.5 Nm — the same order. Sensor noise is the one scenario where the two disagree, and it is the scenario the reference study names as the test of complementarity.

Table 10 crosses the two injection protocols with the presence of the friction term, five seeds each. The result is unambiguous: the fusion margin exists under the inherited protocol in both friction conditions and vanishes under the physical protocol in both.

Table 10. Fusion Margin Under Two Injection Protocols And Two Residual Definitions (PR-AUC, five seeds, mean $\pm$ s.d.).

Configuration	Residual	Raw	Fusion	Margin
Inherited / no friction	0.620 $\pm$ 0.007	0.668 $\pm$ 0.009	0.829 $\pm$ 0.007	+0.161 $\pm$ 0.005
Inherited / friction	0.653 $\pm$ 0.009	0.668 $\pm$ 0.009	0.857 $\pm$ 0.009	+0.189 $\pm$ 0.009
Physical / no friction	0.750 $\pm$ 0.017	0.668 $\pm$ 0.009	0.747 $\pm$ 0.016	-0.003 $\pm$ 0.001
Physical / friction	0.730 $\pm$ 0.014	0.668 $\pm$ 0.009	0.727 $\pm$ 0.014	-0.003 $\pm$ 0.001

Margin is the fused PR-AUC minus the better of the two single models. The raw column is constant by construction, which is also a check that the ablation is controlled. Under the inherited protocol the margin is positive in all five seeds of both rows; under the physical protocol it is -0.003 in all five seeds of both rows, that is, the five per cent raw weight costs a little and returns nothing. Complementarity at window level collapses in the same direction: residual-only detections fall from 27.6 % to 4.8 % and raw-only from 28.0 % to 0.1 %.

The same table answers a second question. Adding the friction term improves the residual model under the inherited protocol by +0.034 $\pm$ 0.009 PR-AUC, in the same direction in all five seeds, but under the physical protocol the change is -0.020 $\pm$ 0.027, which is not significant and points the other way. The friction term's demonstrable benefit is therefore physical fidelity — the 87 % and 92 % reduction of wrist residual spread in Table 3, and fault amplitudes that remain interpretable in newton metres — not detection of these four synthetic faults. It also carries a measured cost: the seed-to-seed spread of the operating threshold rises from 17 % to 46 % of its mean in both protocols.

None of this says the raw model is useless. It says that these four synthetic scenarios, once injected consistently, do not discriminate between the two representation spaces. Section 4.7 reports what the two models did during real events on hardware, which is the only evidence here that bears on real faults.

4.5. Where The Earlier Figures Came From

This subsection collects, in one place, the comparison with the study this work extends. Rebuilt in the mixed-unit configuration of Section 3.8, the pipeline reproduces the

earlier figures point by point (Table 11), which identifies the arrangement they were produced under and therefore the physical scale on which they should be read.

Table 11. Reproduction Of The Earlier Pipeline.

Quantity	Rebuild	Ref.	Δ
Test windows	179,888	179,896	8
Fault windows	14,401 (8.01 %)	14,402 (8.0 %)	1
Residual-only share	23.9 %	24.1 %	0.2 pt
Sensor noise — residual	0.267	0.272	0.005
OR recall	0.841	0.854	0.013
Recall — raw	0.601	0.613	0.012
Encoder glitch — residual	0.975	0.986	0.011
Raw model AUC	0.943	0.952	0.009
Fusion MAX — AUC	0.974	0.976	0.002

The decisive row is sensor noise. A residual model scoring 0.267 on that scenario can only arise on a pipeline where the residual has lost its physical meaning, that is where amperes and newton metres are subtracted directly. Agreement of the window counts to 8 and 1 windows in 179,896 confirms that the same raw dataset is being split the same way.

The structural claim of the earlier study is unaffected. Complementarity under its own injection protocol is measured on both pipelines — 23.9 % residual-only on the reproduction and 27.6 % on the extended one



— and the fusion margin survives on both. What changes is the physical scale on which the numbers should be read, and, separately, what happens to that margin once the faults are injected physically (Section 4.4). Establishing the scale is what made the rest of this work possible: fault amplitudes in newton metres, a friction term with meaningful coefficients, and a residual whose magnitude can be compared with a real contact force all require it.

4.6. Latency And Compute Budget

Per sample, the inverse dynamics evaluation costs 22 μ s and the Jacobian about 30 μ s, together 1.5 % of the 500 Hz budget. Per decision, the two ONNX forward passes cost 5.9 ms, that is 12 % of the 50 ms budget, measured on CPU in offline replay, and 8.5 ms (17 % of the budget) on the cell, where the ROS 2 executor and the logging layer share the same cores; on the deployment workstation with the CUDA provider both fall further. The detection latency is the sum of the 50 ms filter delay and the 50 ms decision period, plus one more decision period when the two-consecutive rule is active, giving 120–150 ms end to end. At a tool speed of 0.25 m/s that corresponds to about 3.75 cm of additional travel.

An end-to-end replay of the prepared data through the deployed class, over the five test runs long enough to carry a fault (27,135 samples, 0.9 minutes of robot time), gives a false-alarm rate of 0.1 % on clean data. Collision and sensor noise are detected in 5 of

5 runs with median latencies of 60 ms and 70 ms; motor drift in 2 of 5 at 2,905 ms; the encoder glitch in none. The latency split is structural — within a 200 ms window a slow ramp is almost a constant offset, and an autoencoder reconstructs a constant offset without difficulty — but the encoder-glitch result is not a latency effect. It is the physical-injection outcome of Table 8 seen at the decision level: a 1.5 rad step on joint 5 propagates into the residual through the Jacobian rather than as the hand-chosen 8 Nm of the inherited protocol, and the propagated disturbance does not cross the threshold. The sample is small and is reported as such: five runs is what the run-disjoint test split contains at that length.

4.7. Commissioning On The Real Robot

All results up to this point were obtained on the recorded dataset. The detector was then commissioned on the physical UR10e cell, covering the two production use cases of the platform — the autonomous visual inspection cycle and a pick-and-place scenario — together with runs in which anomalies were provoked deliberately. Every decision was written to disk, every alarm block to an event log, and every session to a provenance record naming the model checksums, the thresholds and the residual definition in force. That record is not bookkeeping: a first attempt ran with superseded models because the software package shipped them as defaults, and nothing else would have revealed it. Table 12 summarises how the online system deviates from the reference method.

Table 12. Deviations Of The Online System From The Reference Method.

Deviation	Status	Reason
D1 Normalisation bounds and threshold measured on the cell	forced	offline values do not transfer; measured below
D2 Operating threshold for the fused score	forced	the reference method reports Best F1 only, which needs the labels it predicts
D3 Adaptive median + k-MAD rule	optional, off	helps offline, catches no verified event on hardware
D4 Two consecutive decisions before alarm	optional, on	suppresses single-decision noise triggers

D3 and D4 are switched off by a single parameter each, which restores the reference behaviour exactly. D1 keeps the published min-max structure and changes only the set the quantities are estimated from.

The offline threshold does not transfer.

The fused threshold derived from clean validation windows by the rule of Section 3.6 is $\theta = 0.855$, while the median fused score of a clean run on the cell is 0.97: 58.2 % of all decisions crossed it. The scores are not anomalous in scale — under the cell-calibrated configuration of Table 13 the same clean operation sits between 1.7 and 35.3

times below the threshold — so the failure is located unambiguously in where the threshold comes from, not in the models.

The cause is that the models are far outside their training distribution during ordinary operation. Measured against the mean reconstruction error of the offline validation set, the residual model runs at eleven times



that value on the cell and the raw model at eighty-six times. Expressed against each model's own P97 threshold the asymmetry is sharper: the residual model sits at 0.93 of its threshold, which is well calibrated, while the raw model sits at 7.28. The consequence is a distortion of the fusion itself. With $w_{\text{raw}} = 0.05$ the raw model was contributing 29 % of the fused score at the median and dominating it at peaks, because its normalising constant was twenty-five times too small for the cell. Re-estimating both scales on the cell (Section 3.11) returns its share to 8-15 %, in line with its weight.

One further correction was forced by measurement. The min-max form of the reference formula, applied to real-cell data,

locks the residual span to 30.8 because the calibration run contains a payload-carrying phase — an 85-second plateau of fused score whose interior barely moves (median 14.20, p90 14.39, maximum 14.5). The inverse dynamics has no payload model, so carrying anything produces a sustained bias, and a min-max scale fitted over it makes ordinary events invisible. The 97th percentile of each model's clean scores is used instead.

With the cell-calibrated configuration — residual span 25.92, raw span 22.37, $\theta = 5.0$, estimated from 15,627 clean decisions spanning both tasks — the detector produced no alarm across 14.3 minutes of ordinary operation (Table 13).

Table 13. Clean Operation Under The Cell-Calibrated Configuration.

Run	Duration	Median	Maximum	Margin to θ	Alarms
Inspection cycle	6.2 min	0.055	1.28	3.9×	0
Provoked hand contact	1.3 min	0.023	0.14	35.3×	0
Pick and place	6.9 min	0.075	2.92	1.7×	0

Margin is θ divided by the run's peak. The second row is a provoked anomaly that the detector did not register; it is discussed below and is the working limit of the system.

Collisions are detected. In a session of three runs totalling 10.0 minutes, two collisions were provoked against the vacuum gripper. Three alarm blocks were raised; the

operator labelled all three as genuine through the interface, and no unlabelled block occurred. Table 14 lists them and Figure 5 shows the operator interface immediately after the strongest one.

Table 14. Provoked Collisions On The Physical Cell, All Operator-Confirmed.

Event	Duration	Peak	Entry	Trigger	Residual	Raw
1	3.65 s	11.30	6.63	both	170.1 (124×0)	175.9 (447×0)
2	0.25 s	16.13	14.60	both	369.8 (270×0)	466.5 (1186×0)
3	44.95 s	25.04	13.75	both	348.9 (254×0)	427.7 (1087×0)

Residual and raw columns give the model's own reconstruction error and its ratio to that model's P97 threshold. Both models crossed their own thresholds on all three events, and all three were raised by the absolute rule with the adaptive rule disabled. Events 2 and 3 occur seven seconds apart at almost the same pose; the 44.95-second duration of the third is the robot remaining in contact, not a detection that failed to clear.



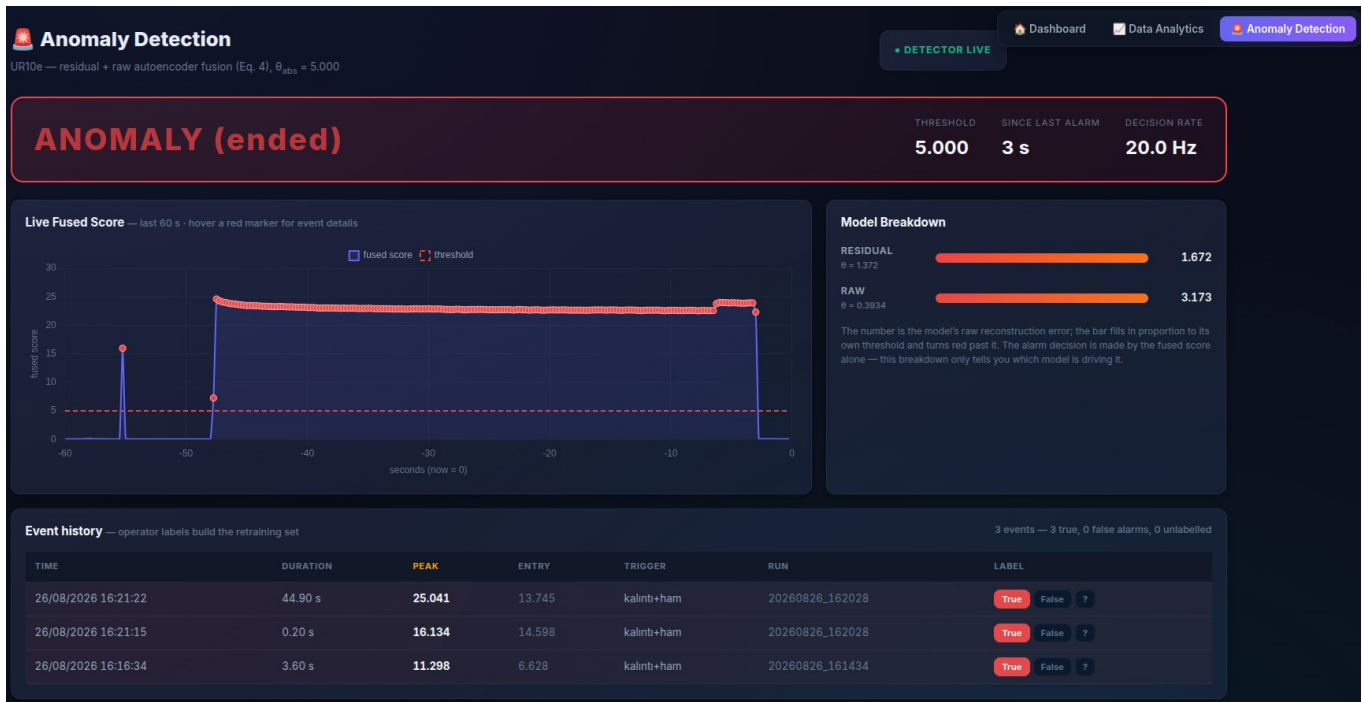


Figure 5. Operator Interface After A Provoked Collision the 44.95-second block of event 3 is the wide plateau at 25; the narrow spike to its left is event 2, seven seconds earlier. The threshold line sits at 5. The breakdown panel gives each model's reconstruction error against its own threshold, and the event table shows all three blocks confirmed by the operator with no unlabelled block. Replayed from the logged decisions of the session.

The separation is genuine rather than a threshold artefact. The weakest confirmed event peaks at 11.30 while the highest excursion in any clean run of that day reaches 4.14, so any threshold in the interval (4.14, 11.30) gives three detections and no false alarm. The deployed value of 5.0 sits at the lower end of that interval, deliberately, to leave margin for events weaker than those observed. Table 15 shows how the interval closes.

Table 15. Threshold Sweep Recounted From All Logged Decisions.

Threshold	Clean blocks / min	Confirmed events caught
3.0	0.00	3 / 3 (fragmented)
5.0 (deployed)	0.00	3 / 3
8.0	0.00	3 / 3
11.3	0.00	2 / 3
16.0	0.00	1 / 3
26.0	0.00	0 / 3

Recounted over 14.3 minutes of clean operation and 10.0 minutes containing the confirmed events, under the same two-consecutive-decision rule the node applies.

A low-amplitude contact is not detected, and no threshold recovers it. The hand contact provoked in the second row of Table 13 raised the fused score to 6.8 times that run's own median, while the clean inspection

cycle routinely reaches 26.7 times its own median. Expressed on the common scale, the peak of the provoked contact falls at the 75.6th percentile of ordinary inspection motion: it is smaller than three quarters of what the robot does normally. Four decision rules were evaluated against this event on the logged decisions — an absolute threshold on two different normalisations, the adaptive median + k·MAD rule swept over $k \in [4, 32]$, and subtraction of a rolling median over windows of 10, 30 and 60 seconds. In every case the clean inspection run produced at least as many blocks as the run containing the event. The signal required to separate them is not present in these two scores.

The working envelope of the deployed system is therefore explicit: events of collision magnitude are detected reliably, with peaks of 11 to 25 against a threshold of 5 and a separation factor of 3.9 over ordinary operation; slow low-amplitude contact is not.

4.8. Operator Labelling

The three events of Table 14 were classified through the interface of Figure 5 while the operator stood at the cell, and the classification is what makes the false-alarm count of Section 4.7 a measurement rather than an impression: the interface reports three events, three confirmed, zero false



alarms and zero unlabelled. Labels are written atomically to a file the detector never reads, so a label can never influence a decision. Together with the per-session provenance record they make a logged session re-analysable months later, which is what allowed the threshold failure of Section 4.7 to be attributed to configuration rather than to the models.

The breakdown panel in Figure 5 also records which model drove each decision. On all three confirmed collisions both models exceeded their own thresholds, the residual model by factors of 124 to 270 and the raw model by 447 to 1186. Together with the undetected hand contact of Section 4.7 this is the only evidence here on how the two spaces behave during genuine faults, and it does not resolve the question raised in Section 4.4: events this large saturate both models, so they cannot discriminate between them, and the one event small enough to discriminate was seen by neither.

5. Discussion

5.1. What The Extension Changed Offline

The extended pipeline scores below the earlier study on the absolute metrics, and the reasons are measurable rather than mysterious. Fault amplitudes are now on the true physical scale, so a ramp specified as 15 Nm is injected as 15 Nm rather than added as a bare number to a channel expressed in amperes. The dataset was cleared of discontinuities that a reconstruction model reads as anomalous. And the split is run-disjoint, so the evaluation no longer measures partly on windows the models had already reconstructed: the clean-window score ratio between test and training runs is 0.48, against the factor of seventy that the earlier arrangement produced.

What survives the extension is the structure. The fusion still exceeds both single models and every baseline; the weight sweep is still broad and flat over the same interval; the complementarity ratios remain close to those of the earlier study, 27.6 % against 24.1 % residual-only and 28.0 % against 25.9 % raw-only. Reporting five seeds rather than one also shows where the remaining uncertainty lives: the ranking metrics vary by less than 0.01 across seeds, while the operating-point precision varies by ± 0.086 , because the threshold is a percentile of a validation distribution narrowed by the split design rather than by the models.

5.2. The Fusion Margin Is A Property Of The Injection Protocol

The central offline finding of this paper is negative, and it concerns the framework's own evaluation rather than its architecture. Under the inherited injection protocol the fusion gains $+0.189 \pm 0.009$ PR-AUC over the better single model, positive in every seed. Under a protocol in which the fault is applied only to the measured channels and the residual is recomputed by the pipeline, the same gain is -0.003 ± 0.001 , also in every seed. Adding or removing the friction term does not change this in either direction (Table 10).

The mechanism is a single amplitude. The inherited protocol perturbs the residual space with a hand-chosen number, and for sensor noise that number is 0.08 Nm, described in the reference study as deliberately small so that the scenario affects the raw signal only. But the extrinsic residual is $r_{\text{ext}} = J(q)^T F$, a linear map of the wrench; it does not attenuate wrench noise, it transforms it. Propagated physically, 3.5 N of wrench noise produces 4.1 Nm of residual disturbance — fifty-one times the injected amplitude — and the residual model detects it perfectly. The complementarity that motivates the fusion was not tested by that scenario; it was defined by it.

This should be read narrowly. The collision scenario's hand-chosen residual amplitude of 40 Nm agrees with the physical propagation, whose peak is 80.5 Nm, so the reference study's choices are not uniformly arbitrary. And the finding does not show that a raw-signal model is useless — it shows that four analytic perturbations, once applied consistently, do not discriminate between these two representation spaces. The



evidence that bears on real faults is in Section 4.7, where both models crossed their own thresholds on all three confirmed collisions.

5.3. Why The Offline Threshold Does Not Transfer

A threshold placed at the 97th percentile of an offline validation distribution declared most decisions on hardware to be alarms. The scores themselves are on an ordinary scale — under the cell-calibrated configuration the same clean operation sits between 1.7 and 35.3 times below the threshold (Table 13) — so the fault is in the estimator, not in the models.

The measurement that explains it is the distance of the models from their training distribution during ordinary operation: eleven times the validation mean error for the residual model and eighty-six times for the raw model. A percentile of a distribution the deployment never occupies carries no information about the deployment. The asymmetry between the two models is the practically important part. The residual model, whose input has had the trajectory-dependent dynamics removed by the physical model, sits at 0.93 of its own threshold on the cell — well calibrated. The raw model, reading the signals directly, sits at 7.28. With a five per cent weight it was nevertheless contributing 29 % of the fused score, because a normalising constant twenty-five times too small turns a small weight into a large influence. This is a general trap in score-level fusion: the weights mean what they say only while the normalisers are valid.

The split design itself contributes, and this is the trade-off at the centre of the extension. To keep validation and test inside the training joint range — necessary, or a high score on the cell would mean 'unseen pose' rather than 'anomaly' — the sixteen validation runs were selected for similarity to training. Validation loss falls accordingly, to 0.036 for the raw model against 0.320 in the reference study, and with it the width of the distribution θ is drawn from. The same choice that removed the leakage therefore produced a threshold too narrow for the cell. This is not a defect of the models: their loss is identical with and without the friction term, so it cannot be attributed to the residual definition either. It is a property of estimating an operating point on data selected to resemble training.

The remedy used here keeps the published formula and changes only the set the quantities are estimated from: clean decisions logged on the cell, spanning every task the cell performs, with a robust percentile in place of the min-max range. The last detail is not cosmetic. Applied to real-cell data the min-max range is set by an unmodelled payload — the inverse dynamics has no payload term, so carrying a workpiece produces a sustained residual bias, an 85-second plateau whose interior varies by 0.3 — and a span fitted over it makes ordinary events invisible.

5.4. What The Friction Term Bought

Adding a Coulomb-plus-viscous term outside the solver removes 87 % and 92 % of the residual spread on wrist_2 and wrist_3, the two channels where the inverse dynamics previously contributed almost nothing and the residual was in practice the raw measurement. That is a real gain in physical fidelity and it makes fault amplitudes interpretable in newton metres on all six joints.

It is not a detection gain. Under the inherited protocol the residual model improves by $+0.034 \pm 0.009$ PR-AUC, consistent across seeds; under the physical protocol the change is -0.020 ± 0.027 , not significant. It also costs threshold stability: the seed-to-seed spread of the operating threshold rises from 17 % to 46 % of its mean. The honest summary is that the term corrects the model, and that correcting the model does not by itself improve the detection of these particular synthetic faults. It is retained in the deployed configuration because real faults are not these four, and because a residual that means what it says is easier to reason about when something unexpected happens.

5.5. The Working Envelope

The deployed detector catches collisions and does not catch slow low-amplitude contact, and both halves of that sentence are measured. Three operator-confirmed collisions peaked between 11.30 and 25.04 against a threshold of 5.0, with no unlabelled alarm block in ten minutes and no alarm at all in fourteen minutes of ordinary operation; any threshold between 4.14 and 11.30 gives the same result. A deliberately provoked hand contact, by contrast, raised the fused score to 6.8 times that run's median while ordinary inspection motion routinely reaches 26.7



times its own — the provoked event sits at the 75.6th percentile of normal operation. Four decision rules were evaluated against it and none separated it.

The limit is therefore not the threshold but the representation. An event smaller than the variation that normal motion already produces cannot be isolated by any monotone rule on these two scores. Closing that gap needs either a model trained on the cell's own trajectories, so that normal operation stops being out-of-distribution, or an additional channel that responds to contact rather than to dynamics. Stating the envelope is more useful than a single headline number: a safety function whose sensitivity floor is unknown is harder to deploy than one whose floor is published.

5.6. Limitations

- A single robot (UR10e) and a single task profile; no validation on other robot types has been performed.
- Synthetic faults, even injected physically, remain analytic perturbations. The collision pulse is distributed over the wrench channels rather than propagated from a real contact point, and the four scenarios were shown in Section 4.4 not to discriminate between the two representation spaces once injected consistently.
- The current-to-torque coefficient could be measured directly on only two of six joints; the other four are derived by assumption, which affects the physical interpretation rather than the detection.
- The friction coefficients are fitted by a regression that absorbs any velocity-correlated model error, not friction alone. No independent tribological validation was performed.
- The inverse dynamics has no payload model. Carrying a workpiece produces a sustained residual bias that the deployed threshold sits above but does not explain away.
- The operating threshold is measured on the cell, which is necessary, but it is measured on a calibration run of 15,627 decisions and validated on twenty-four minutes of operation. The envelope it defines is a commissioning-scale result with respect to thermal drift, wear, part-to-part variation and operator differences.

- Only three confirmed anomalous events were available for the deployed configuration. The threshold interval (4.14, 11.30) is bounded below by observed clean operation and above by the weakest of those three events; a weaker real event would narrow it.
- Online measurements were taken on CPU; they should be repeated with the GPU provider on the deployment workstation.

6. Conclusions

This paper took a published score-level fusion framework for cobot anomaly detection, extended it, and delivered it as a detector running on a physical UR10e cell. Every measurement taken on the way is reported, including the ones that are unfavourable to the framework.

The starting framework was rebuilt independently and its architectural and procedural components match exactly. Establishing the physical scale of the signals was a precondition for everything that follows: the driver writes motor current, not torque, into the effort field, so the earlier figures were produced on a mixed-unit pipeline. That pipeline was reproduced to within 0.013 on every quantity, which fixes the scale on which the earlier numbers should be read and makes fault amplitudes, friction coefficients and residual magnitudes physically meaningful for the extension.

On that basis the pipeline was extended. A friction term absent from the validated solver was added outside it, removing 87 % and 92 % of the residual spread on the two wrist joints where the inverse dynamics had previously contributed almost nothing, and the evaluation was placed on a run-disjoint split with five training seeds. Under the inherited injection protocol the fusion retains its advantage: $F1\ 0.791 \pm 0.023$ against 0.627 ± 0.006 and 0.614 ± 0.016 for the single models, a gain of +0.164, with complementarity ratios close to those of the earlier study (27.6 % against 24.1 %).

The injection protocol itself was then examined, and this is the paper's principal negative result. When the same four faults are injected only into the measured channels and the residual is recomputed by the pipeline, the fusion margin falls from $+0.189 \pm 0.009$ to -0.003 ± 0.001 PR-AUC, in every seed and



irrespective of the friction term. The mechanism is a single amplitude: the reference protocol perturbs the residual space for the sensor-noise scenario with 0.08 Nm, chosen so that the scenario affects the raw signal only, whereas the physical propagation of the same 3.5 N wrench disturbance through $r_{\text{ext}} = J(q)^T F$ produces 4.1 Nm. The scenario the reference study names as the test of complementarity does not test it. The complementarity of the two representation spaces is therefore reported here as a property of the evaluation protocol, not of the models.

The system was implemented as a ROS 2 node running at 500 Hz whose feature engine is numerically identical to the offline pipeline to floating-point rounding, friction term included, in which both models run on every decision and whose inference consumes 12 % of the decision budget.

On hardware the offline-derived threshold did not transfer, declaring 58 % of decisions to be alarms. The cause was measured rather than assumed: during ordinary operation the residual model runs at eleven times its validation mean error and the raw model at eighty-six times, so a percentile of the validation distribution carries no information about the deployment. The asymmetry is the practically useful part — the residual model sits at 0.93 of its own threshold on the cell while the raw model sits at 7.28, which let a five-per-cent-weighted model supply 29 % of the fused score. Re-estimating both scales and the threshold on clean cell decisions, with a robust percentile in place of the min-max range, restores the intended weighting.

With that configuration the detector was validated against provoked collisions. Three operator-confirmed events were caught with peaks of 11.30, 16.13 and 25.04 against a threshold of 5.0, with no unlabelled alarm block in ten minutes of event-bearing operation and no alarm at all in fourteen minutes of ordinary operation; any threshold in (4.14, 11.30) gives the same result. The working envelope is bounded on the other side too: a deliberately provoked low-amplitude hand contact reached only 6.8 times its own run median while ordinary inspection motion reaches 26.7 times, placing it at the 75.6th percentile of normal operation. Four decision rules were evaluated against that event and none separated it.

Two directions follow. The sensitivity floor is a property of the representation, not of the threshold, so closing it requires models trained on the cell's own trajectories — which would also remove the out-of-distribution condition behind the threshold-transfer failure — or an additional channel that responds to contact rather than to dynamics. The second is a payload term outside the solver, following the same pattern as the friction term, since the unmodelled payload is what forces the normalising scale up during transport. Both are supported by the instrumentation already in place: every decision, every alarm block, the operator's label and the provenance of the configuration that produced them are written to disk.

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Contribution Of Researchers

Author 1: design and implementation of the offline pipeline, the audit and the ROS 2 detector node, commissioning measurements, preparation of the manuscript. Author 2: Author 3: (To be completed by the authors.)

Conflict Of Interest

No conflict of interest has been declared by the authors.

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